0.5 Project Scope Sheet · Group 4

Draft for discussion · 13 November 2026 · T. Santos, L. Batista

Ship and Ocean Systems Design 2026 · Ship Design Project · Scope sheet for your group

Field	Value
Group	Group 4
Members (student number)	96720 · 105343 · 121344
Baseline	A · Azores Express: container ship, 1,300 TEU, dual-fuel HFO/LNG, Lisbon – Ponta Delgada – Praia da Vitória – Lisbon, weekly
Specialization 1	SP05 · Parametric Hull Shape Generation
Specialization 2	SP11 · Advanced Dimensional Analysis
Deadline	Dec 13, 2026, one ZIP file by e-mail to luis@veranavis.com
ZIP name	Group4_Project.zip
Presentation	Dec 18, 2026, 15 minutes per group

Revisions

All alterations to the project scope of work and/or design requirements for the group, against 0.2 Ship Design Project · Baseline A, Azores Express: Owner's Requirements (p. 4), are recorded in the following table:

Rev.	Date	Description	By
0	To be set	First issue, after discussion with the students	T. Santos, L. Batista

Baseline A

Deliverable Groups D1–D8 and D10

ID	Deliverable	Status
D1	**Design Basis.** Every requirement tagged Given / Assumed / To be checked against 0.2 Ship Design Project · Baseline A, Azores Express: Owner's Requirements (p. 4)	
STU-D1.1	Owner's requirements and port limits: ship type and target capacity, 1,300 TEU at 10 t per TEU on average (13,000 t cargo) · weekly round voyage Lisbon – Ponta Delgada – Praia da Vitória – Lisbon · sequence of port calls (round trip) · relevant characteristics and limits at Lisbon, Ponta Delgada and Praia da Vitória (and Leixões), each with its source	
STU-D1.2	Laws and regulations applicable, incl. environmental targets	
STU-D1.3	Operating speeds: service 18–19 kn, trial and top speed	
STU-D1.4	LNG bunkering concept: LNG bunkered in mainland Portugal · number of round voyages between bunkerings as a design decision of the group, stated and justified	
D2	**Reference Ship Database.**	
STU-D2.1	Reference ship database with sources: at least 10 gearless container ships of about 900–1,800 TEU, recent	
STU-D2.2	Regressions and bounds: analysis of the statistical data, as in Appendix A.3	
D3	**Ship Model.** Inputs, outputs, sequence and extent of the calculations (volume carrier, Table 1.9) · estimation method selected and named for every item, within its range of validity · outputs that respond correctly to changes in the inputs (qualitative check)	
STU-D3.1	Ship model (computer codes and/or files): hull form coefficients (C_B, C_P, C_M, C_WP, vertical prismatic) · tonnage (GT, NT)	

ID	Deliverable	Status
STU-D3.2	Weight estimation, assumptions and margins: lightship weight (steel, outfit, dual-fuel machinery, LNG storage aft and superstructure, with centres) · deadweight components (cargo, LNG and HFO, lubricating oil, fresh water, provisions and stores, crew) · closed weight balance	
STU-D3.3	Power and propulsion: propulsive coefficients (wake, thrust deduction, relative-rotative and hull efficiency) · propulsive power by Holtrop & Mennen, dual-fuel machinery, single shaft · open-water propeller efficiency η0 as a stated, justified assumption (0.60) · propeller selection with a cavitation check · side thruster · electric power estimate for each operating condition · fuel blend as a design assumption: share of the voyage in gas and in diesel mode, and the pilot fuel	
STU-D3.4	Subdivision and capacities: subdivision and compartments (bulkheads, engine room and peaks) · capacities of holds and tanks · LNG storage sizing: LNG per round voyage × round voyages between bunkerings → net volume, then the filling limit, the tank type and the space around it → gross volume; tank type and location as design decisions of the group · TEU lost where the LNG storage sits · container capacity sizing (technical note) and TEU capacity check, containers in the holds and on deck	
D4	**Voyage and Financial Model.** One round voyage a week · ship model outputs as input · the LNG solution carried into the costs	
STU-D4.1	Voyage model: sea and port time · LNG, pilot fuel and HFO use, by the fuel blend of D3.3 · bunkering calls · fuel and port costs · freight revenue · global results for one round trip and for one year of operation	
STU-D4.2	Financial model: building cost, incl. the extra cost of the dual-fuel plant and the LNG system · financing, RFR and NPV · EU ETS and FuelEU Maritime: whether they apply to the route under the rules in force, and a future scenario with tighter environmental regulation	
D5	**Validation Note.**	
STU-D5.1	Validation against reference ships: your model against the reference ships, with the choice justified · error per quantity (L, B, T, Δ, lightship weight, propulsive and electric generation power, newbuilding cost) · model corrected where needed	
STU-D5.2	Hold utilization k_U: calibration on the reference ships and range, used in D6	
D6	**Optimization Report.** Minimum propulsion power that keeps the weekly schedule · trade-off between LNG storage volume and bunkering interval (TEU lost against bunkering calls), in total cost and operations	
STU-D6.1	Optimization report against the design objective, sensitivity analysis: problem definition (design variables, restrictions with their sources, justified objective(s)) · exploration of the domain of the variables · active constraints · optimum against the requirements · delivered code that reproduces the results presented	
STU-D6.2	Code flowchart (ISO 5807): a flowchart of your code, matching it, that shows how the information flows (inputs, each calculation step, the iterations and the outputs), in the symbols of ISO 5807 used correctly	
D7	**Compliance Sheet.** Each applicable rule identified with its source, and each result shown against its limit	
STU-D7.1	EEDI and CII as design targets, with the LNG carbon factor and the methane slip stated	
STU-D7.2	Preliminary intact stability assessment: requirements and typical conditions, incl. ballast conditions · stability with the deck stacks	
STU-D7.3	Draught at every port	
STU-D7.4	Freeboard	
STU-D7.5	Emission control area near the mainland: Tier III NOx, 0.10% sulphur	
STU-D7.6	LNG fuel system safety (IGF Code): tank location, bunkering and gas-safe machinery spaces	
D8	**General Arrangement and Hull Form.**	
STU-D8.1	Hydrostatics check: final main dimensions and hydrostatic properties (displacement, C_B) against the targets, with the error levels discussed	
DWG-D8.1	General arrangement (A3): profile, midship section and main deck, with the main dimensions and the cargo area configuration (bays, rows and tiers), the forward superstructure with the bridge sight line, and the LNG storage aft, with the tank connection space, the bunkering station and the vent mast	

ID	Deliverable	Status
STU-D8.2	Container count on your own lines: final cargo capacity (TEU, deadweight) against the targets, with the error level discussed	
STU-D8.3	3D hull form, justified: 3D hull shape with the containers in the holds and on deck, the main cargo-hold bulkheads, the collision bulkhead and the deckhouse exterior (files)	
D10	**Datasheet and Presentation.**	
STU-D10.1	Datasheet (template 0.4): one page, A4	
STU-D10.2	Presentation, 15 minutes: oral presentation of the results in PowerPoint, including: objectives of the work (description of the initial concept of the ship); computation model (the computation process, with the adopted methods, criteria and validations); assessment of the results obtained (comparison with existing similar ships); conclusions	
STU-D10.3	Report (0.3 Report Template): a clear description of all the work, a summary of the resulting ship characteristics and a comparison with the statistical data of existing ships of the same market segment	

Specialization scope

One report per specialization goes in D9, with the drawings listed. Each specialization has its own self-study deck.

Deliverable Group D9

ID	Deliverable	Status
D9	**Specialization Documents (2).** One individual report per specialization, with every required study and drawing (below) · methods and assumptions stated · results checked against the acceptance criteria of the specialization	

SP05 · Parametric Hull Shape Generation

Generate your hull from a parent hull, the KCS (Bibliography, *Hull geometry*), and make it a variable of your optimization model: the Lackenby section shift sets the fullness and LCB, and the stretch sets L, B, T and D (the reading Hull Form Variation from a Parent and Demo 8, under Tools and resources).

ID	SP05 deliverable	Status
STU-SP05.1	Hull generator: the KCS offsets checked against the published particulars (∇, C_B, LCB); Lackenby and the stretch (including depth), with the check that each hull hits its target C_B and LCB and that ∇ = C_B L B T	
STU-SP05.2	Hull family: hulls across C_B and LCB (and L/B, B/T) inside the validity range; for each hull, C_P, C_M, C_WP, KB and BM from the hull, and the effective power by Holtrop–Mennen with the hull's own coefficients; the Holtrop check against the KCS model test stated	
STU-SP05.3	Capacity on the hull: bay-by-bay count (Stage A of Lecture 3.2) on each generated hull, holds and deck separately; the counted k_U compared with 0.5 C_B + 0.4	
STU-SP05.4	Coupling to the optimizer: C_B, LCB, L, B, T and D as variables of the model; the counted TEU, ∇, P_E, KB and BM returned to D3–D6 in place of the formulas; the effect on the optimum compared with Baseline A	
STU-SP05.5	Limits: the validity range used (C_B, L/B, B/T), the bulb and stern only stretched, the Holtrop uncertainty, and where the chosen design sits in that range	
DWG-SP05.1	Lines plan of the chosen hull (body plan, waterlines, sectional area curve) and a 3D view, with the parent and the transformation stated	

SP11 · Advanced Dimensional Analysis

Build relationships from data: dimensionless groups first, then regression, then a small neural network, and judge which one to trust.

The SP11 report is a data-modelling report, justifying:

ID	SP11 deliverable	Status
STU-SP11.1	Dimensional analysis of two quantities of your ship model (e.g. propulsion power and lightship or steel weight): the governing variables with their units, the dimensionless groups by the Buckingham Π theorem, and the choice of groups justified	
STU-SP11.2	Regressions in dimensionless groups on the reference ships of D2 (power laws, fitted in log space), against the plain regressions of STU-D2.2: fit quality (R^2, residuals), number of coefficients and range of validity	
STU-SP11.3	Dataset generated with your ship model of D3 (or with Holtrop–Mennen): sampling over the design ranges, inputs as dimensionless groups, split into training and test rows	
STU-SP11.4	Neural network with one hidden layer, built in Excel cells and trained with Solver, then the same network in Python; number of neurons and inputs justified	
STU-SP11.5	Comparison of the network with a linear and a quadratic regression on the test rows, inside and outside the training range · which model to use in D6, and why	